TÉCNICO LISBOA

AVTM 2026/2027

Exercise 3 (week 3, September 23th and September 25th)

The tasks of exercise are:

1. Incorporate graphics elements with transparency in your application.

2. Implement the Fog effect.

3. Insert a view of the rear-view cam (camera looking backwards) by using the stencil mechanism. See the attached AVT_StencilDemo.zip for an example.

4. Allow to pause the game when the user presses a key ('P' key), showing a message while paused. Implement the counting mechanism of lives, points and the endgame. We suggest the following game play: at the beginning the player has five lives and lose a life each time the car is hit by an orange or falls off the table. When the player loses all lives, you should see a game over message and it should be possible to start a new game by pressing a button ('R' key). Implement a HUD (Head-up Display) to show the remaining lives as well as the number of won points. It is intended that the display information of points and lives is carried out through a Head-Up Display (HUD) which involves the use of a second orthogonal projection to the drawing. This projection will be independent of the active camera. In this case it may be worth doing push and pop the stack projection matrices. You can base your implementation on the code provided in the myAVTM_Template which uses TrueType library which uses bitmap fonts for text rendering: https://dev.to/shreyaspranav/how-to-render-truetype-fonts-in-opengl-using-stbtruetypeh-1p5k

5. Improve the graphic quality of your application by using the Assimp library to load OBJ file format objects. You can use the code from AVT_assimp_demo.zip. Just download the zip file, decompress it and open the Solution in the VS 2026. Run it in the x64 platform. Then, in the console enter a directory name containing the OBJ file; for instance: backpack.